

# Back to Square One

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Foundations of AI

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# How this works

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- Nine questions. Some are multiple choice: put a hand up.
- The rest you **write down yourself**, on paper or on a laptop, and then we □ll them in together on the screen.
- Everything on these slides was on a slide today.

A question the room gets wrong is worth □ve it gets right, so guess out loud and be wrong cheerfully. That has been the whole day.

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*Section 1*

# The loop

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# 1. Write the algorithm

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Block 1, one house.  $x$  is its size in square metres,  $y$  the price it sold for, and  $w$  the price per square metre we are looking for. The model is  $y = w * x$ .

```
def fit(x, y, w, alpha):  
    _____ # the verdict: y minus the estimate  
    _____ # while more than half a franc out  
        _____ # the update: move w the way the miss points  
        _____ # score the new guess  
    return w
```

**Your turn.** Write the four lines, then we'll put them in together.

```
def fit ( x, y, alpha, tol )
```

```
    w = 200
```

```
    while abs(loss) > tol :
```

```
        y_hat = w * x
```

```
        loss = y - y_hat
```

```
        w = w + alpha * loss
```

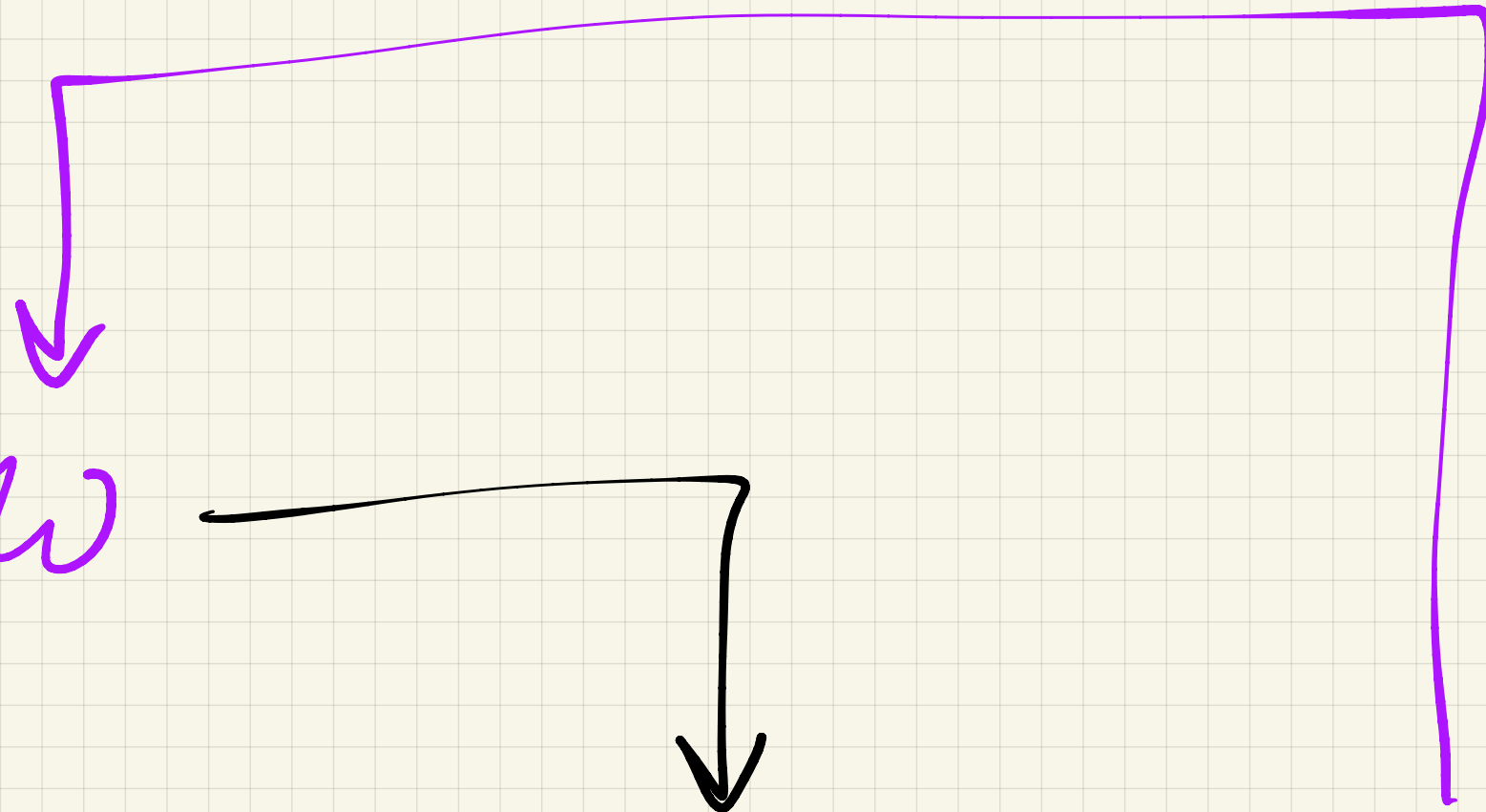
$y$   
price

$x$   
area

$$x \longrightarrow \hat{y} = wx \longrightarrow l = y - \hat{y}$$

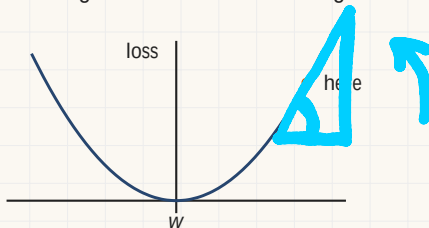
$w$

$$w = w + \alpha l$$



## 2. The tilt, on the side of the valley

The curve is the loss. The dot is where the guess is now. What is the gradient there?



- A) about  $-12$
- B) exactly 0
- C) about 0.1
- D) about  $+12$**

### 3. The tilt, at the bottom

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Same curve. The dot has moved.



- A)** about  $-12$
- B)** exactly  $0$
- C)** about  $+0.5$
- D)** about  $+12$

## 4. Which of these can draw any shape at all?

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Given enough of them. More than one answer is right.

- A)** Neural networks
- B)** Straight lines
- C)** Polynomials
- D)** Loss functions

## 5. The network inside a language model

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You type half a sentence and press enter. What does the network itself hand back?

- A)** A word
- B)** A matrix of words
- C)** One probability for every word it knows
- D)** The word with the highest score

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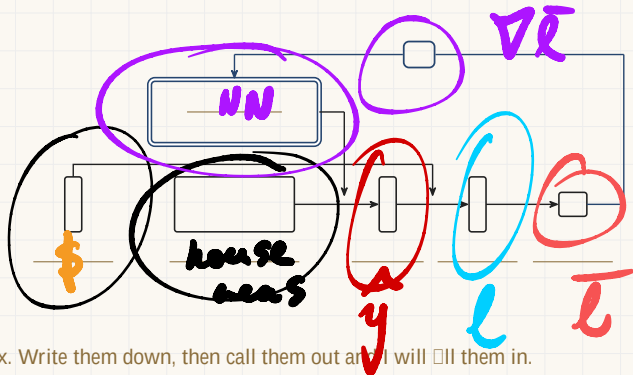
*Section 2*

# Name the boxes

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## 6. The same ☐ ve boxes: house prices

Every application today was this one picture. Here it is a network that predicts house prices.



**Your turn.** Name all six. Write them down, then call them out and I will ☐ll them in.



## 8. That wider box, on its own

Two short texts, the cat sat and the dog ran. The five words along the top are everything this toy model knows.

y x  
"cat" "the"  
"sat" "the cat"  
"ran" "the dog"

| the | cat | sat | dog | ran |
|-----|-----|-----|-----|-----|
| 0   | 1   | 0   | 0   | 0   |
| 0   | 0   | 1   | 0   | 0   |
| 0   | 0   | 0   | 0   | 1   |

after *the*, the next word was cat

after *the cat*, it was sat

after *the dog*, it was ran

**Your turn.** Three things. What is one **row**? What is one **column**? And if the model were perfect, what would be in every **cell**?

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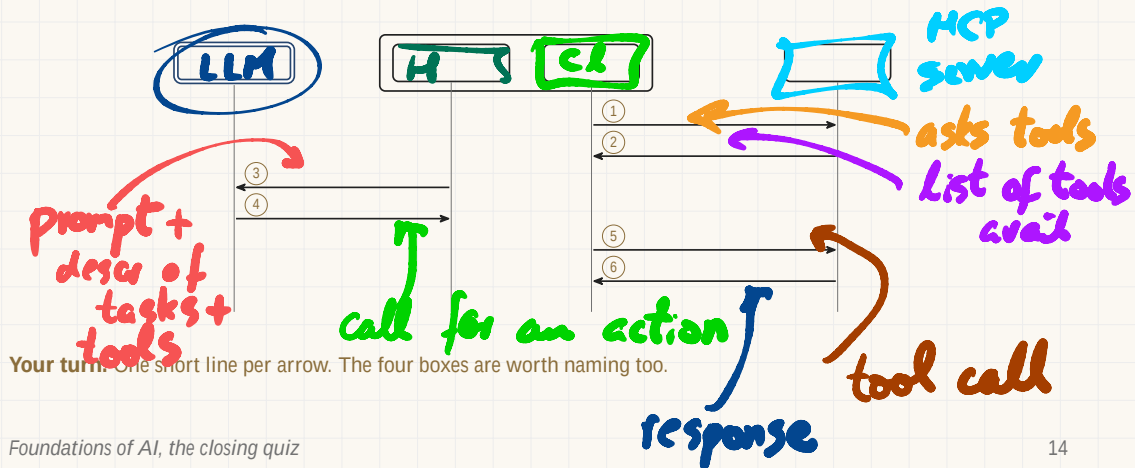
*Section 3*

# The agent

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## 9. What travels along each arrow?

The four actors of this afternoon, and the six messages between them, in order.



# That is the day

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- A guess, a measurement of how wrong it was, and a correction. Nine o'clock to five.
- The same five boxes behind a house price and behind a sentence.
- The only honest number comes from data the model never saw.
- A model writes text. A harness does things. Govern the harness.

Your project is on the course page: about an hour, in the browser, and it builds the house price network you met at question 6. Thank you for a Saturday.